COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

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Annexure A

SHORT ANSWER TEST

Code of Conduct:

I D. Sheshuraaj (196526004) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Concepts	25%	1	1	1	1
Explanation	25%	1	2	1	1
Application	20%	1	1	2	1
Organization	15%	1	1	1	1
Accuracy	10%	1	1	1	2
Clarity	5%	2	1	1	1

Total Marks 29 /40

Signature of the Faculty

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Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of

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3 MB each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10.A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation

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Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organizatio n	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professiona l presentatio n	Understanda ble with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

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CO1 - AT1

28

CSA0709
Computer Networks
Dr. Shankar R

1. A company needs to transfer a 150 MB design file from a client computer to a remote server over a network with a bandwidth of 50 Mbps. The transport layer divides the file into 1500-byte segments, while the data Link layer encapsulates each segment into frames by adding 40 bytes of header and trailer. Calculate the total number of segments and frames transmitted. The total transmission overhead introduced by the Data Link layer, and estimate the total transmission time. Explain how the transport and Data link layers ensure reliable connection.

2. A transport layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the total number of segments. If one frame in every window is lost and must be retransmitted, determine the total number of retransmission. Explain how flow control improves communication efficiency.

3. A 12,000-byte IP packet is transmitted through a router where the maximum transmission Unit (MTU) is 1500 bytes. Assuming each fragment requires a 20-byte IP header, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network layer ensures successful delivery of fragmented packets to the destination.

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4. A transport layer protocol uses a sliding window size of 6 frames. Each frames carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmission. Explain how to flow control improves communication efficiency.

1. No of segments:

150 MB = 150×10^6 bytes

Segments = $(150 \times 10^6) / 1500$ = 100,000 segments

No of frames:

Assuming one frame per segment = 100,000 frames

Data link overload:

Overload = $100,000 \times 40$ = 4,000,000 bytes = 4MB

Transmission time:

Total transmitted bytes = 150,000,000 + 4,000,000
= 154,000,000 bytes

Total bits = $154,000,000 \times 8$ = 1,232,000,000 bits

Time = $1,232,000,000 / 50,000,000$ = 24.64 seconds

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Layer functions:

Transport layer: -segments data and may use sequence numbers, acknowledgements and retransmissions depending on the protocol.

Data Link layer: frames data, provides link addressing, error detection and may retransmit damaged frames on supported links.

2. Given:

Window size = 6 frames, user data per frame = 1024
total frame = 48.

Number of windows:

Windows = 48/6 = 8 frames

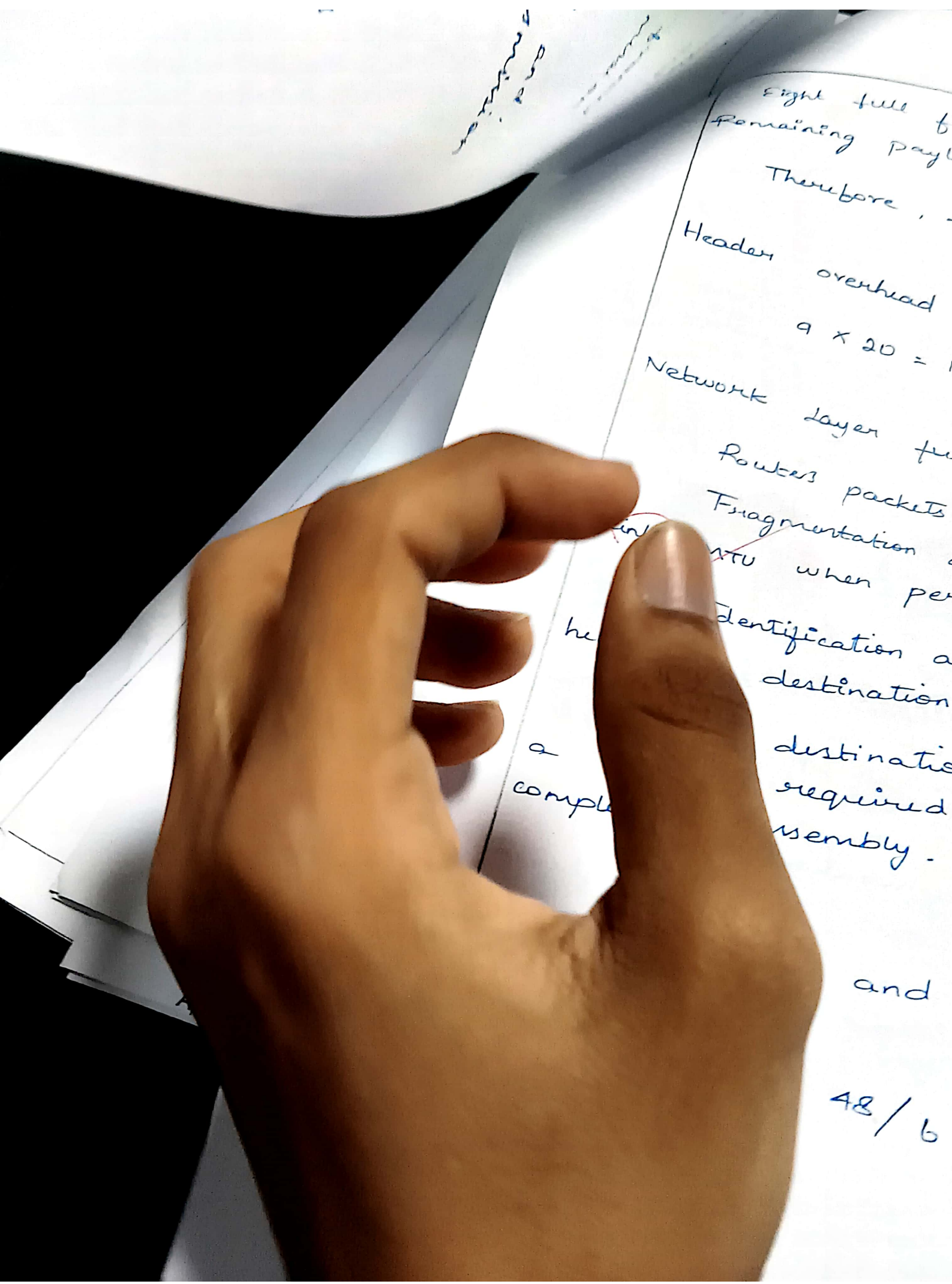
Retransmission:

One lost frame per second = 8 x 1 = 8 re

Total user data:

48 X 1024 = 49,152 bytes

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Flow control :

Allows multiple frames to be sent before waiting for acknolegment.

Helps prevent reciever buffer overflow.

Improves bandwidth utilization and reduces unnecessary waiting.

Supports recovery from lost frames through acknowlegments and retransmission when the protocol supports them.

Conclusion : Eight windows are required and, under the given assumption, eight transmission occur.

3. Given:

Original IP packet = 12,000 bytes, MTU = 1500 bytes
IP header per fragment = 20 bytes.

Maximum data per full size fragment

1500 - 20 = 1480 bytes

Number of fragments :

Assuming 12,000 bytes includes the original 20-byte IP header, original payload = 12000 - 20 = 11,980 bytes.

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Eight full fragments carry $8 \times 1480 = 11840$ bytes

Remaining payload $= 11980 - 11840 = 140$ bytes

Therefore, total fragments = 9 fragments

Header overhead :

$9 \times 20 = 180$ bytes

Network layer functions :

Routes packets using logical IP addresses

Fragmentation allows a packet to fit a link MTU when permitted.

Identification and fragment offset fields help the destination associate fragments.

The destination reassembles fragments, a missing required fragment can prevent complete reassembly.

Sliding Window and flow control :

Windows $= 48/6 = 8$ windows

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Retransmission :

8 windows × 1 lost frame = 8 retransmissions

Total user data :

5. Given :

File size = 600 MB , checkpoint interval = 60MB , failure occurs after 390 MB

Checkpoint before failure :

390/60 = 6 checkpoints at 60, 120, 180, 240, 300 and 360 MB.

Data to retransmit :

Last completed checkpoint = 360 MB

390 - 360 = 30MB must be retransmitted after recovery , assuming checkpoint state is preserved.

Importance of Synchronization :

Manages communication sessions between applications.

Creates synchronisation points during long transfers.

Helps resume from saved progress after failure.

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